VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	23BAT1112

Continuous Assessment Test (CAT) – II - OCTOBER 2025

Programme	:	B.Tech. CSE with all specializations	Semester	:	Fall Semester 2025-26
Course Code & Course Title	:	BCSE305L- Embedded Systems	Class Number	:	CH2025260102123 CH2025260102124 CH2025260102125 CH2025260102126 CH2025260102127
Faculty	:	Dr. Kishor Kisan Ingle Dr. Parvathy A K Dr. Sankar P Dr. Premanand V Dr. Sudharson S	Slot	:	C2+CT2
Duration	:	90 minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Description	Marks	CO	BT Level
1	Rani is an image-sensor designer who created a tiny Limpel-Ziv lossless compression module from short numerical status-strings formed only from the symbol set A {3,4,5,6}. **a.** What encoded output and built dictionary can she expect for the uncompressed text "3535353453536636"? **(5 marks)** **b.** Also, provide the number of binary bits required to store the encoded output? **(2 Marks)** **c.** Instead, if she had used the symbol set B {7,8,9,0} and got 9,0,29,38,8,7,27,19 as the compressed output, what decoded text and built dictionary could she expect? **(3 Marks)**	10	2	4
2	A team is building a smart water meter with a microcontroller. They use an interpreter for rapid prototyping. For production, they switch to a compiled and assembled binary executable file to optimize the speed and memory. Finally, they load the program using a device programmer. a) Compare the trade-offs between using an interpreter Vs a compiler-assembler in this context. **(2 Marks)** b) Why is a device programmer preferred here over in-system programming (ISP)? Illustrate the key differences between them. **(5 Marks)** c) If the program needs to be updated remotely after deployment, which programming method would you choose, and why? **(3 Marks)**	10	1	4

3	a) For the following code snippet draw a control data flow graph. **(5 marks)**
	b) Also provide significance of different nodes of control data flow graph. **(5 marks)**
	`int foo(int x) {`
	`  int result = 0;`
	`  if (x > 0) {`
	`  for (int i = 0; i < x; i++) {`
	`  if (i % 2 == 0)`
	`  result += i;`
	`  else`
	`  result -= i;`
	`  }`
	`  } else {`
	`  result = -1;`
	`  }`
	`  return result;`
	`}`
		10	3	4

4	Consider the following code segment:
	`for (int i = 0; i < n; i++) {`
	`  int x = 5 * 4;`
	`  int y = i * 16;`
	`  int z = (a + b) - (a + b);`
	`  sum = sum + (x * 1) + (y / 2) + (c * d) - (c * d);`
	`}`
	a. Identify two different types of code optimization that are relevant to this example. **(4 Marks)**
	b. Apply at least three optimization techniques to rewrite the optimized code. **(4 Marks)**
	c. Explain how these optimizations improve the execution efficiency of the program. **(2 Marks)**
		10	3	4

5	A real-time embedded control system executes the following periodic tasks: • **Task T1:** C = 2 ms, P = 5 ms, D = 5 ms • **Task T2:** C = 3 ms, P = 10 ms, D = 7 ms • **Task T3:** C = 4 ms, P = 20 ms, D = 15 ms a) Using Rate Monotonic Scheduling (RMS), analyze whether the task set is schedulable for the given deadlines. **(5 marks)** b) If the task set fails under RMS, evaluate whether Earliest Deadline First (EDF) can guarantee all deadlines. Justify your answer with calculations. **(5 marks)**	10	4	4

**********All the best **********

Faculty Signature